Year 10 Foundation

REVISION PACK

Algebra 1

Name: _____

Class: _____

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: substitution

$a = 5$ and $b = -3$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 4b$

(a)	**Step 1**	Put the numbers in	$2 \times 5 + (-3)$
	Step 2	Answer	7
(b)	**Step 1**	Put the numbers in	$5^2 - 4 \times (-3)$
	Step 2	Power and × first, then add	$25 + 12 = 37$

Practice: substitution

1.1 $a = 4$ and $b = -1$

(a) Work out the value of $3a + b$ (b) Work out the value of $a^2 - 2b$

1.2 $a = 2$ and $b = -5$

(a) Work out the value of $5a + b$ (b) Work out the value of $a^2 - 1b$

1.3 $a = 6$ and $b = -2$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 3b$

1.4 $a = 5$ and $b = -3$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 4b$

1.5 $a = 3$ and $b = -4$

(a) Work out the value of $5a + b$ (b) Work out the value of $a^2 - 2b$

1.6 $a = 7$ and $b = -1$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 5b$

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: simplifying

Simplify

(a) $6a + 3b - 2a + b$ (b) $5x \times 2y$

(a)	**Step 1**	Collect a	$6a - 2a = 4a$
	Step 2	Collect b	$3b + b = 4b$
	Step 3	Answer	$4a + 4b$
(b)	**Step 1**	Multiply the numbers	$5 \times 2 = 10$
	Step 2	Answer	$10xy$

Practice: simplifying

1.7 Simplify

(a) $4x + 5y - x + 2y$ (b) $6a \times 3b$

1.8 Simplify

(a) $7p - 2q - 3p + 6q$ (b) $2c \times 7d$

1.9 Simplify

(a) $8a + b - 5a + 4b$ (b) $5m \times 4n$

1.10 Simplify

(a) $9x + 2y - 4x + 3y$ (b) $4p \times 6q$

1.11 Simplify

(a) $3m + 8n - m - 5n$ (b) $7x \times 2y$

1.12 Simplify

(a) $6c + d - 2c + 4d$ (b) $3a \times 8b$

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: expanding brackets

Expand

(a) $3(4x - 2)$ (b) $x(x + 7)$

(a)	**Step 1**	Multiply $4x$ by 3	$3 \times 4x = 12x$
	Step 2	Multiply -2 by 3	$3 \times (-2) = -6$
	Step 3	Answer	$12x - 6$
(b)	**Step 1**	Multiply x by x	$x \times x = x^2$
	Step 2	Multiply 7 by x	$x \times 7 = 7x$
	Step 3	Answer	$x^2 + 7x$

Practice: expanding brackets

1.13 Expand

(a) $4(3x - 5)$ (b) $x(x + 6)$

1.14 Expand

(a) $6(2x + 1)$ (b) $x(x - 3)$

1.15 Expand

(a) $2(5x - 4)$ (b) $x(x + 9)$

1.16 Expand

(a) $7(2x - 3)$ (b) $x(x + 8)$

1.17 Expand

(a) $5(3x + 1)$ (b) $x(x - 2)$

1.18 Expand

(a) $3(4x - 7)$ (b) $x(x + 1)$

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: multiplying the same letter

Simplify $4p \times 3p$

Step 1 Multiply the numbers	$4 \times 3 = 12$
Step 2 $p \times p$	p^2
Step 3 Answer	$12p^2$

Practice: multiplying the same letter

1.19 Simplify $3m \times 5m$

1.20 Simplify $4x \times 2x$

1.21 Simplify $6y \times y$

1.22 Simplify $5a \times 4a$

1.23 Simplify $2k \times 9k$

1.24 Simplify $7n \times 3n$

Algebra 1 · Solving equations (Lesson 2.4)

Worked example: two-step equations

Solve $4x + 7 = 31$

Step 1	Subtract 7	$4x = 24$
Step 2	Divide by 4	$x = 6$

Practice: two-step equations

2.1 Solve $5x + 3 = 38$

2.2 Solve $2x + 9 = 25$

2.3 Solve $6x - 4 = 32$

2.4 Solve $4x + 7 = 35$

2.5 Solve $7x - 2 = 40$

2.6 Solve $3x + 8 = 29$

Algebra 1 · Solving equations (Lesson 2.4)

Worked example: equations with brackets

Solve $3(x-4)=18$

Step 1	Expand	$3x-12=18$
Step 2	Add 12	$3x=30$
Step 3	Divide by 3	$x=10$

Practice: equations with brackets

2.7 Solve $5(x-3)=20$

2.8 Solve $2(x+6)=22$

2.9 Solve $4(x-1)=28$

2.10 Solve $3(x+5)=27$

2.11 Solve $6(x-2)=30$

2.12 Solve $2(x-7)=10$

Algebra 1 · Solving equations (Lesson 2.4)

Worked example: unknowns on both sides

Solve $6x - 5 = 2x + 15$

Step 1	Subtract $2x$	$4x - 5 = 15$
Step 2	Add 5	$4x = 20$
Step 3	Divide by 4	$x = 5$

Practice: unknowns on both sides

2.13 Solve $5x - 2 = 2x + 13$

2.14 Solve $8x + 1 = 3x + 26$

2.15 Solve $9x - 7 = 4x + 8$

2.16 Solve $6x + 4 = 2x + 24$

2.17 Solve $7x - 5 = 3x + 11$

2.18 Solve $5x + 3 = 2x + 18$

Worked example: equations with x over a number

Solve $\frac{x}{4} + 5 = 8$

Step 1	Subtract 5	$\frac{x}{4} = 3$
Step 2	Multiply by 4	$x = 12$

Practice: equations with x over a number

2.19 Solve $\frac{x}{4} + 2 = 7$

2.20 Solve $\frac{x}{5} + 3 = 9$

2.21 Solve $\frac{x}{2} + 6 = 13$

2.22 Solve $\frac{x}{6} + 1 = 4$

2.23 Solve $\frac{x}{3} + 7 = 12$

2.24 Solve $\frac{x}{5} - 2 = 4$